Reconstructing the full bilocal fluctuation

A strong Volterra limit and explicit spectral amplitudes

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Abstract

The previously constructed joint finite-time memory kernel is now connected to the original singlet bilocal. A Hermitian dilation of the energy-pinched memory makes its background propagator contractive, including degenerate energy blocks. Interpolation between the collective second-moment bound and the existing force eighth moment controls the return terms, giving an $O(p^{-1/3})$ strong reconstruction error on fixed time intervals. The charged spectral amplitude simplifies to a single centered Hermite coefficient; the neutral amplitude is a regular second divided difference of the exponential. Their common positive spectral matrix gives the complete bilocal covariance. This is a result for the specified microscopic model; the bulk mass and position map remains to be constructed.

1 Model and the actual singlet observable

Use standard CAR and the overlapping complex Gaussian p-body Hamiltonian, with the same original disorder across charge sectors. The state is normalized annealed full-Fock trace at $\beta = \mu = 0$. Take even N, $p \to \infty$, $\lambda_N = p^2/N \to \lambda > 0$, fixed J_s, $\hbar = 1$, and string time. Write

$$C = J_s^2/\lambda, \quad q = e^{-4\lambda}, \quad b_{N,m} = 2^{-N}\binom{N}{m}. \tag{1}$$

In a sector $m > 0$, put $h = H_m$, $K = I_N \otimes H_{m-1}$ and

$$V\psi = m^{-1/2}\sum_i |i\rangle c_i \psi, \quad P = VV^\dagger, \quad R = (1-P)KV, \quad D = (1-P)K(1-P). \tag{2}$$

On $P \oplus (1-P)$, K has system block αh, where $\alpha = 1 - p/m$. Its compressed propagator and memory are

$$F(t) = V^\dagger e^{iKt}V, \quad M(t) = R^\dagger e^{iDt}R, \quad F' = i\alpha hF - M * F, \quad F(0) = 1. \tag{3}$$

Convolution is ordered: $(A * B)(t) = \int_0^t A(t-s)B(s)ds$. The center-time-zero bilocal and its dynamic part are

$$S_c(t) = me^{-iht/2}F(t)e^{-iht/2}, \quad \mathcal{B}_N(t) = (1-\mathcal{P}_h)S_c(t), \quad \mathcal{P}_h Y = \sum_e P_e Y P_e. \tag{4}$$

The sum keeps whole energy eigenspaces. Inactive $m < p$ sectors have zero dynamic field, including $m = 0$. We reconstruct this unnormalized fluctuation, whose limit is nontrivial even though $S_c/m \to 1$ [1].

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2 A contractive background and an exact difference

Set $M_0 = \mathcal{P}_h M$ and $\delta M = M - M_0$. Let F_0 solve the same Volterra equation as F, with M_0 in place of M. It is not enough to assume that a pinched memory has a stable propagator; here this follows from an explicit dilation.

For every energy projector P_e, introduce a separate copy of the original bath and define

$$R_0\psi = (RP_e\psi)_e, \qquad D_0 = \bigoplus_e D, \qquad K_0 = \begin{pmatrix} \alpha h & R_0^\dagger \\ R_0 & D_0 \end{pmatrix}. \tag{5}$$

Then $R_0^\dagger e^{iD_0 t} R_0 = M_0(t)$. Consequently F_0 is the system compression of $e^{iK_0 t}$ and

$$\|F(t)\|_\infty \le 1, \qquad \|F_0(t)\|_\infty \le 1. \tag{6}$$

Each system energy block interacts only with its own bath copy. Thus F_0 commutes with h, but need not be scalar inside a degenerate block. Kernels $M_0(s)$ and $M_0(t)$ need not commute.

For $\operatorname{Re} z > 0$, the compressed resolvents are

$$\widehat{F} = (z - i\alpha h + \widehat{M})^{-1}, \qquad \widehat{F}_0 = (z - i\alpha h + \widehat{M}_0)^{-1}. \tag{7}$$

The resolvent identity gives the exact ordered relations

$$F - F_0 = -F_0 * \delta M * F, \tag{8}$$

$$= -F_0 * \delta M * F_0 + F_0 * \delta M * F_0 * \delta M * F. \tag{9}$$

No infinite perturbation expansion is used. The preceding memory field is

$$\mathcal{F}_N(s) = (1 - \mathcal{P}_h) m e^{-ihs/2} M(s) e^{-ihs/2}, \qquad Q_N(s) := m\delta M(s) = e^{ihs/2} \mathcal{F}_N(s) e^{ihs/2}. \tag{10}$$

The already proved uniform bound is

$$\sup_N \sup_{|s| \le S} \|Q_N(s)\|_{2,\mathrm{ann}} < \infty \quad (S < \infty). \tag{11}$$

In particular (8) identifies the natural scale without discarding the finite return term. It also implies $\|m(F - F_0)(t)\|_2 \le t^2 \sup_{s \le t} \|Q_N(s)\|_2/2$.

3 Which norm is being used

For sector operators let

$$\|X\|_{r,A}^r = \sum_{m \in A} b_{N,m} \mathbb{E} \operatorname{tr}_m |X_m|^r. \tag{12}$$

The Gaussian expectation and charge sum form one finite trace; Schatten Hölder and interpolation apply in this algebra. Rectangular factors are normalized by the same source dimension. All estimates below are uniform on a fixed real-time interval; constants can depend on that interval and a positive compact interval containing λ_N.

4 Interpolation controls the collective return

The existing force square-function estimate [2] gives, for $G = R^\dagger R$ and $\mathcal{A}_m = \sum_i [H, c_i]^\dagger [H, c_i]|_m$,

$$0 \le G \le \mathcal{A}_m/m, \tag{13}$$

$$\mathbb{E}\,\mathrm{tr}_m\, \mathcal{A}_m^k \le (2k-1)!!(Nv_N)^k, \qquad Nv_N = O(p). \tag{14}$$

Trace eigenvalue monotonicity, not operator monotonicity of the power map, gives $\mathbb{E}\,\mathrm{tr}_m\, G^k \le m^{-k}\mathbb{E}\,\mathrm{tr}_m\, \mathcal{A}_m^k$. Rectangular Hölder yields

$$\|M(s)\|_{k,m,\mathrm{ann}} \le \|G\|_{k,m,\mathrm{ann}} \le ((2k-1)!!)^{1/k} Nv_N/m. \tag{15}$$

Pinching contracts every Schatten norm, so M_0 has the same bound and δM at most twice this bound.

Take the good charge set $A_N = \{m \ge N/4\}$. On it, the stronger second moment follows from (11), whereas the eighth moment follows from (14):

$$\|\delta M(s)\|_{2,A_N} = O(p^{-2}), \qquad \|\delta M(s)\|_{8,A_N} = O(p^{-1}). \tag{16}$$

Thus the standard 2–8 interpolation inequality gives

$$\|\delta M(s)\|_{4,A_N} \le \|\delta M(s)\|_{2,A_N}^{1/3} \|\delta M(s)\|_{8,A_N}^{2/3} = O(p^{-4/3}). \tag{17}$$

This does not assert a uniform fourth moment for $m\delta M$.

Let $U(t) = e^{iht}$ and $U_\alpha(t) = e^{i\alpha ht}$. Using (6), the exact integral equations $F = U_\alpha - U_\alpha * M * F$ and $F_0 = U_\alpha - U_\alpha * M_0 * F_0$ imply

$$\|F - U\|_{4,A_N} + \|F_0 - U\|_{4,A_N} = O_S(p^{-1}). \tag{18}$$

Indeed M, M_0 have fourth norm $O(p^{-1})$, while $\|(\alpha - 1)h\|_{4,A_N} \le (4p/N)\|h\|_4 = O(p^{-1})$. The uniform fourth energy moment is already available in this model.

In the exact first line of (8), the integrand error after replacing both propagators by U is

$$(F_0 - U)\, m\delta M\, F + U\, m\delta M\, (F - U). \tag{19}$$

Hölder, $m \le N = O(p^2)$ and (17)–(18) bound its second norm on A_N by $O(p^{-1/3})$. On the remaining active set $p \le m < N/4$,

$$\sup_{|s|\le S} \|Q_N(s)\|_{2,A_N^c} \le Kp\, e^{-N/16}, \tag{20}$$

using the sector bound $\|m\delta M\|_2 \le 2\sqrt{3}\, Nv_N$ and $\sum_{m<N/4} b_{N,m} \le e^{-N/8}$. There the propagator differences have operator norm at most two. This treats all charges without a new four-source moment theorem.

5 Strong reconstruction of the entire bilocal

Define, for $t \geq 0$,

$$(\mathcal{V}_h \mathcal{F}_N)(t) = -\int_0^t ds \int_{-(t-s)/2}^{(t-s)/2} du\, e^{ihu} \mathcal{F}_N(s) e^{-ihu}. \tag{21}$$

For negative times use the adjoint relation. This is the whole memory field, with its charged, neutral and mixed components retained.

Theorem 1. *For each fixed $S < \infty$, in the stated model and limit,*

$$\sup_{|t| \leq S} \|\mathcal{B}_N(t) - (\mathcal{V}_h \mathcal{F}_N)(t)\|_{2,\mathrm{ann}} \leq K_S p^{-1/3} + K_S p e^{-N/16}. \tag{22}$$

Both fields have uniformly bounded second norms on this interval.

Proof. The background $m e^{-iht/2} F_0(t) e^{-iht/2}$ is removed exactly by $1 - \mathcal{P}_h$. Apply the preceding integrand bounds to (8), whose two-time simplex has area $t^2/2$. After replacing its propagators by U, put their two times equal to $a, b \geq 0$, with $a + b = t - s$. Equation (10) combines all phases into $e^{ihu} \mathcal{F}_N(s) e^{-ihu}$, where $u = (a - b)/2$ and $da = du$. This gives (21) with no extra factor. Its pinching is zero already, since $\mathcal{P}_h(e^{ihu} \mathcal{F}_N(s) e^{-ihu}) = 0$. Conjugation preserves the second norm, so $\|\mathcal{V}_h \mathcal{F}_N(t)\|_2 \leq t^2 \sup_{s \leq t} \|\mathcal{F}_N(s)\|_2/2$. The uniform bound and (22) complete the proof. □

Consequently the two covariance functions differ by $O_S(p^{-1/3})$ uniformly in all external real center times T. This is a comparison of two finite-N covariances. Their limit below is asserted for fixed T; no growing-time spectral limit is taken.

Let $\mathcal{K}_{\mathcal{F}}(T; s, r)$ be the complete memory covariance from [3]. Dominated convergence and (22) give

$$\mathcal{K}_{\mathcal{B}}(T; t, v) = \int_0^t ds \int_0^v dr \int_{-(t-s)/2}^{(t-s)/2} du \int_{-(v-r)/2}^{(v-r)/2} dw\, \mathcal{K}_{\mathcal{F}}(T + w - u; s, r). \tag{23}$$

Here $t, v \geq 0$, with other signs determined by adjoints. The old finite-time memory construction has thus been connected to the actual unnormalized singlet fluctuation, rather than only to its first few Taylor coefficients.

6 Explicit charged and neutral spectral amplitudes

Recall the common positive two-energy matrix ν_{ab} for $\mathcal{X}_a \in \{J_d\ (d \geq 2), N_j\ (j \geq 1)\}$ from [3]. Here N_j denotes a neutral residual field, not fermion number. Energies E, E' label input/column and output/row. Put

$$e = (E + E')/2, \quad \omega = (E' - E)/2, \quad \chi_L(E' - E) = \frac{\sin(\omega L)}{\omega}, \quad \chi_L(0) = L. \tag{24}$$

With $P_d = [d]_q!$ the monic Hermite coefficients are

$$c_d(s) = \frac{\int e^{isx} H_d^{(q,C)}(x) d\rho_q(x)}{C^d P_d}. \tag{25}$$

The centered memory amplitudes are $e^{-ies} D_d(s; E, E')$ and $2Ce^{-ies} c_j(s)$, where $D_d = -c_d'' + 2iec_d' + EE'c_d$. Therefore (21) gives

$$b_a(t; E, E') = -\int_0^t \chi_{t-s}(E' - E) e^{-ies} f_a(s; E, E') ds. \tag{26}$$

For the charged part let $g_d(s) = e^{-ies} c_d(s)$. Direct differentiation gives $e^{-ies} D_d = -g_d'' - \omega^2 g_d$. The Green-function identity is

$$-\int_0^t \chi_{t-s} e^{-ies} D_d(s) ds = g_d(t) - \cos(\omega t) g_d(0) - \chi_t g_d'(0). \tag{27}$$

Since $c_d(0) = c_d'(0) = 0$ for $d \geq 2$, the charged amplitude is simply

$$\boxed{b_{J_d}(t; E, E') = e^{-iet} c_d(t).} \tag{28}$$

The zero initial data are essential; this identity is not asserted for the discarded $d = 0, 1$ channels.

For the neutral part use the standard second divided difference of $f_t(x) = e^{itx}$ [4]. Integrating the scalar Green function, or checking its initial-value equation, gives

$$\boxed{b_{N_j}(t; E, E') = \frac{2Ce^{-iet}}{C^j P_j} \int H_j^{(q,C)}(x) [E, E', x] f_t \, d\rho_q(x).} \tag{29}$$

The apparent denominators in the distinct-node interpolation formula are removable. A definition valid also for coincident real nodes is

$$[x_0, x_1, x_2] f_t = (it)^2 \int_{a,b \geq 0,\ a+b \leq 1} e^{it[ax_0 + bx_1 + (1-a-b)x_2]} da \, db. \tag{30}$$

Thus $|[x_0, x_1, x_2] f_t| \leq t^2/2$, and at three coincident nodes it equals $-t^2 e^{itx_0}/2$. Two applications of the fundamental theorem of calculus establish the integral identity, so no complex mean-value point is assumed.

7 The resulting full covariance and its first coefficients

The complete bilocal answer is

$$\boxed{\mathcal{K}_{\mathcal{B}}(T;t,v)=\sum_{a,b}\int e^{iT(E'-E)}\overline{b_a(t;E,E')}b_b(v;E,E')d\nu_{ab}(E,E').} \tag{31}$$

All charged/neutral mixed entries of the same original-disorder matrix are included. The time and field sums can be integrated termwise: on $0\le t\le S$, $|\chi_{t-s}|\le t-s$, and the previously proved memory tails are absolutely summable in the field Hilbert norm and in spectral total variation. The Volterra map has norm at most $S^2/2$ on these time families. This transfers those tails to (31) without an unproved uniform fourth moment.

Each finite spectral block has an $L^2(\rho_q\otimes\rho_q)$ density. The full time-dependent spectral measure is absolutely continuous by the total-variation limit, and has no diagonal static mass. Only L^1 regularity is claimed for the infinite density. Positivity of ν yields a nonnegative equal-time norm and $\mathcal{K}_{\mathcal{B}}(T;t,v)^*=\mathcal{K}_{\mathcal{B}}(-T;v,t)$.

Writing $\mathcal{F}(s)=A_0+is(A_1+B_1)+O(s^2)$ gives

$$\mathcal{B}(t)=-\frac{t^2}{2}J_2-\frac{it^3}{6}\left[J_3-\frac{3}{2}\{H,J_2\}+2CN_1\right]+O_{L^2}(t^4), \tag{32}$$

$$\mathcal{K}_{\mathcal{B}}(T;t,v)=\frac{t^2v^2}{4}\mathcal{K}_{J_2}(T)+O\big(|t|^3|v|^2+|t|^2|v|^3\big). \tag{33}$$

These are expansions of the established limiting field. They recover the earlier $-t^2Y_-/2$ bilocal curvature. The nonzero memory channel $\nu_{A_0B_2}$ is retained in (31); no independence assumption sets its contribution to zero.

8 What physical information is still needed

This stage determines the dynamic bilocal covariance in the stated infinite-temperature double-scaled model. Its Gram metric can now be tested against candidate bulk constructions. That metric by itself does not supply canonical bosonic dressing or a light-front Hamiltonian. The existing Itakura construction [5] fixes a vacuum, normal ordering, polarization and physical boson subspace before its meson projection. Our finite normalized trace obeys $\mathrm{tr}[A,B]=0$ exactly, so a nonzero vacuum CCR expectation cannot be read off in the same state with convergent commutator moments.

The next task is to identify the relevant algebra and state, and a microscopic spatial/internal-momentum generator. The energy difference $E'-E$, the frequency conjugate to memory time, and a meson's internal quark momentum fraction are presently different variables. A genuine P^- and its interactions must be derived before comparing with the already known bulk mass operator. The present result does not assign particle masses to the spectral frequencies or establish finite-G gravitational four-point dynamics.

9 Independent checks and reproducibility

Two new scientific outputs are exclusive-create and retained unchanged. The final evidence binds their input hashes, parameters and dependency versions, predecessor files, both compilation passes and every final rendered page. No earlier scientific main is rerun.

- Finite Volterra checks: two nontrivial CAR sectors $(N, p, m) = (4, 2, 3), (5, 2, 3)$, one zero-memory $m = p$ control, and an independent system with a two-dimensional degenerate energy block. There are eight Laplace-resolvent cases, sixteen time-domain integral checks, twenty-four contraction checks and twelve curvature values. The maximum second-order resolvent error is 2.69×10^{-15}. Time quadratures use orders 8 and 14. Nonzero finite return terms are saved; these small systems are not a double-scaling fit.

- The degenerate example has noncommuting $M_0(0.3), M_0(0.8)$ (normalized Hilbert–Schmidt commutator about 0.0331). Replacing whole eigenspaces by separate eigenvectors changes the background memory, by about 0.259 in the recorded test. This checks that block degeneracy is treated as part of the definition.

- Spectral-filter checks: 1,260 charged and 1,440 neutral exact rational coefficient identities through degree 14. Two physical λ values, four signed times, ordinary/repeated/near-repeated energy pairs and six field degrees give 144 floating field cases. Independent spectral/time sizes $(40, 24)$ and $(64, 36)$ agree to 2.54×10^{-15}. The time-integral and explicit-amplitude forms differ by at most 2.40×10^{-15}.

- Twenty additional repeated-node cases compare a 3×3 triangular matrix exponential with a simplex integral. The triple-node value and $t^2/2$ bound are checked separately. All floating comparisons retain their errors and are not interval certificates.

The strong-return theorem uses the analytical norm estimates, not extrapolation from these tests. This stage remains local, with no Git publication or external issue reply.

References

[1] A. Day and Chengheng, *The compressed evolution and memory of the complete bilocal*, local project stage `susskind-bilocal-channel` (2026).

[2] A. Day and Chengheng, *Two-return reduction of finite memory*, local project stage `susskind-memory-return-expansion` (2026), force high moments; and `susskind-memory-uniform-bound`, collective second norm.

[3] A. Day and Chengheng, *The joint finite-time memory kernel*, local project stage `susskind-memory-joint-regression` (2026).

[4] NIST Digital Library of Mathematical Functions, Sec. 3.3(iii), Eqs. 3.3.34–3.3.37, divided differences and analytic continuation.

[5] K. Itakura, *Boson Expansion Methods in (1+1)-dimensional Light-Front QCD*, arXiv:hep-th/9604032, Secs. 4–5. The source record distinguishes its light-front conventions and vacuum from the present model.